

CSE341_Assignment1&2

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CSE 341: Microprocessors

Department of Computer Science and Engineering

BRAC University

Theory Assignment 1

1. Calculate the smallest and fourth largest segment for the following physical addresses also write down the logical address for each of them:

a. FB2DEh

b. 18C3Dh

2. Describe pipelining and the function of the pre-fetch queue, also explain the significance of pipelining in 8086.

3. Explain all the tasks of the EU and BIU.

Address	B0000	BF21F	C1234	B7465	B7466
Data	FEH	ABH	47H	24H	98H

4. Given, CS = 1000h, SS = 3011h, DS = B000h, ES = E54Ah, IP = 0001h, SI = 5231h, DI = 7888h, SP = F57Eh, BX =1000h.

a. Calculate the physical address of the current index in the extra segment.

b. Calculate the physical address of the top of the stack.

c. According to the table, determine in which segment the value a)ABH and b)47H is in, explain your

answers with a proper logical address and if you are unable to find the segment for any of these cases, state your reasoning in mathematical form.

d. MOV DX,[BX+SI+1234H], Determine the value in DX.

5. Find out 2 different possible logical addresses for each of the following physical addresses. If it is not possible to find also write your explanation:

a. 10h

b. A2312h

c. Can the physical addresses in (a) and (b) as a segment number?

Theory Assignment 2

1. Explain all the functions of Flag Registers(Status and Control) and why we need them.

2. Find out all the values of flag registers with detailed calculations.

a. MOV AL, F5h

MOV BL, E2h

ADD AX, BX

b. MOV AL, F5h

MOV Ah, E2h

MOV Cl, 95h

MOV Ch,21h

ADD AX, CX

3. Given, CS = 2010h, SS = 4021h, DS = 615Eh, ES = E74Ah, IP = 5001h, SI = 888h, DI = 7888h, SP = F57Eh, BX =1000h, BP = 78h

a. MOV AX,05h

b. MOV [BP], AX

c. MOV [2222h],0F9h

d. MOV CX, [BX+1222h]

e. PUSH AX

f. POP CX

g. MOV DX,CX

h. MOV [DI],DX

i. MOV BX,BX[6000h]SI

j. HLT

Name all the addressing modes, and calculate their physical address(if not possible leave it). Also determine what values will be stored in AX, CX, DX, and BX at the end.

4. Convert these assembly instructions to machine language

a. MOV AL, [BP+1342h]

b. MOV [BX+DI+8211h],DX

5. Convert these machine code to assembly instructions

a. 88160070 h

b. 89801962 h

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